

# A MODEL OF OPTIMAL SCIENCE FUNDING: TARGETING THE CAPABILITY-RESOURCE GAP

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their capabilities and in their resources; research output requires both, and funding compounds capability across rounds. We show that the optimal allocation funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records underdetermine the gap; peer review supplies an additional, noisy signal of capability. We find that this signal’s value increases with capability inequality and with the signal’s informativeness, and that under heavy-tailed capability distributions even a noisy signal from peer review captures most of that value. We further find that uniform seed grants and lotteries, which forgo targeting, lose little output exactly where review is worth little, and that whom to fund matters far more than when to fund.

## 1. INTRODUCTION

Science funding is large and consequential, and how best to allocate it remains unsettled. The United States spent close to a trillion dollars on research and development in 2024, roughly a fifth of it funded by the federal government (??). Whether that money should be concentrated among a few researchers or spread across many is a long-running dispute (?). How well peer review predicts research outcomes is contested: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (?), while studies with other designs find modest but real predictive power (???). And whether review should be replaced, in part, by lotteries has moved from proposal to practice (??).

These disputes concern a single underlying problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. The empirical evidence cannot settle them on its own, because each study measures one input to that decision, how well scores predict outcomes among funded grants or how output

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responds to grant size, and none can say what the funder should do with what it knows. That question depends on what the funder is uncertain about, on what its information is worth, and on how the answer changes with the budget. It is a question for decision theory. This paper treats the funder’s problem as a sequential decision under uncertainty and analyzes it with standard Bayesian methods. It answers four questions: whom to fund, whether to spend early or wait and learn, what information about researchers is worth, and how seed grants and lotteries perform relative to targeted funding. The model is deliberately simple, and its contribution is to our theoretical understanding: it derives the optimal allocation, shows how the three disputes are structured, and identifies the two features of a field on which the answer to each of them turns.

Consider the challenge of the grant funder, whether a federal agency disbursing billions or a private foundation disbursing millions: you have a fixed budget and must choose how to allocate it across researchers and research projects to create the greatest positive impact. Two intuitive schemes might be considered. Fund the track record: give to those who have produced the most, since past productivity is evidence of future productivity. Fund the under-resourced: give to those who have the least, since a dollar goes further where dollars are scarce. Both, we will show, are suboptimal, and for the same reason: each accounts for only one of the two quantities the impact of a grant depends on.

We model this decision situation as follows. A funder with a fixed budget allocates grants, over multiple rounds, to researchers who differ in two ways: epistemic capability, what a researcher knows and can do, and resources, what a researcher has to work with. Research output requires both and is limited by the scarcer of the two; the funder can observe output, but not capability or resources themselves. A grant adds to a researcher’s resources in the round it is given and, because doing research builds expertise, raises their capability in later rounds; we call this compounding. Each round, the funder observes the output that results, updates its beliefs about the researchers, and allocates again.

Our results are as follows. When the funder knows every researcher’s capability and resources, the optimal allocation funds the capability-resource gap: each funded researcher receives a grant increasing in their capability and decreasing in their existing resources, and the budget determines how many researchers receive funding at all. Each intuitive scheme fails because it accounts for only one of the two quantities. In reality the funder lacks direct access to the gap, and track records alone underdetermine it: we show that a funder relying on the track record performs far worse than one that estimates the gap.

Peer review supplies an additional signal of capability. The value of that signal to the funder is the expected output it adds by reducing the funder’s uncertainty about the gap, and that value is greatest where capability is unequally distributed and review informative; in fields with heavy-tailed distributions of capability, even a noisy signal from peer review captures most of this value. The same reasoning says how uniform seed grants and lotteries perform relative to targeted funding: each gives up the value of the targeting it replaces, which is small where review is worth little and large where review is worth most. We vary the budget throughout, so our results apply to funders of every size, and the budget is one of the two features that decide every result: targeting matters most per dollar for a small foundation, whose grants are small relative to researchers’ total resources, and whether optimal funding should be concentrated among a few researchers or spread across many depends on the budget relative to those resources. No funding mechanism is good or bad in itself; the funder’s budget and the field’s distribution of capability decide which mechanism adds output. Finally, when to fund matters far less than whom to fund: the best split of the budget between early and late rounds shifts with the funder’s budget, but the difference is small relative to the difference between funding the right and the wrong researchers.

Two assumptions limit these results. First, grants are used up in the round they are given; whatever lasts from a grant (skills, publications, standing) is in the form of naturally compounding researcher capability generated by the chance to produce more research. Second, capability and resources are complements throughout: neither fully substitutes for the other. We defend each assumption where it is used.

We proceed as follows. §2 situates our model in the literatures on funding allocation, peer review, and lotteries. §3 presents our model. §4 derives the optimal allocation. §5 shows what track records can and cannot reveal about the gap, then measures the value of peer review and relates it to the empirical evidence on review. §6 examines the timing of funding, including the strongest argument for spending early. §7 measures how seed grants and lotteries perform relative to targeted funding and asks how widely funds should be spread. §8 discusses implications and limitations, and §9 concludes.

## 2. RELATED LITERATURE

Our contribution sits at the intersection of four literatures: the empirical debate over concentrating or dispersing funds, the evidence on how well peer review predicts research outcomes, the case for funding lotteries, and formal models of science funding.

Whether funders should concentrate resources on a few researchers or spread them across many is a long-running dispute, and concentration is the current practice: the top 1% of NIH principal investigators receive close to 10% of research project grant funding, a share that has risen since 1985 (??). Empirical studies of funded portfolios generally find diminishing marginal returns to funding per researcher, and reviews of this literature conclude that the evidence favors more spreading than current practice (??). Concentration is also self-reinforcing: early grant winners accumulate advantages beyond the grant itself (?). This dispute typically concerns the returns to funding: whether output per researcher increases or decreases as grants concentrate. In our model, the answer also depends on the size of the funder’s budget relative to researchers’ total resources: with a budget large enough to relieve most bottlenecks, the optimal allocation itself reaches nearly every researcher, so little is lost by spreading funds widely; with a tight budget, effectiveness requires concentrating funds on the researchers with the largest capability-resource gaps (§7). The value of any targeting, in turn, depends on what the funder can know about researchers (§5).

How well peer review predicts research outcomes is contested. Among funded NIH grants, percentile scores predict subsequent productivity barely better than chance (?), and reviewers assessing the same application agree only weakly (?). Studies with other designs find real but modest predictive power: better scores predict more publications, citations, and patents, including under controls for investigator history and institution (?), and a natural experiment in funding finds that higher-scored proposals produce more than lower-scored ones (?). Both sides of this dispute measure review’s accuracy: how well scores predict outcomes among funded applicants. Our model instead measures review’s value: the expected output that review adds by reducing the funder’s uncertainty about the capability-resource gap. That value depends on the field; the same noisy signal can be worth a great deal in one field and nothing in another (§5), and no single number settles the question.

Doubts about review have made lotteries a live proposal. Advocates argue that randomizing among applications that pass a review threshold saves review costs, reduces bias, and honestly reflects review’s limited ability to discriminate among proposals that pass a review threshold (???), and several funders now run partial lotteries (???). The existing formal argument for lotteries measures what a lottery saves: ? model grant competition as a contest in which scientists invest costly effort in proposals, show that when only a small fraction of proposals is funded this effort can rival the scientific value of the research funded, and show that a partial lottery among proposals above a threshold removes the link between the wasted effort

and the fraction funded. What has not been measured is what a lottery gives up: the value of targeting. Our model measures that value. It is negligible where budgets are ample, review is uninformative, or capability is evenly spread, and substantial where capability is heavy-tailed and budgets are tight (§7). Together the two measures let a funder weigh the choice field by field.

Formal models of science funding have concentrated on the incentive costs of competition and on the value of exploration. Beyond the contest model above, ? find, in a game-theoretic model of contests for scarce rewards, that intensifying competition pushes scientists toward higher-risk, higher-return projects. ? simulates funding strategies on an epistemic landscape and finds that allocation with an explicitly random component significantly outperforms a peer-review-like strategy on large landscapes, because review concentrates effort on regions already known to be productive. ? model allocation as a decision under heavy-tailed uncertainty about project value, and combine a citation-based signal with a lottery weighted by that signal. In each of these models, funding raises a single quantity, a project’s chance of success or a researcher’s productivity, and none treats the timing of spending as the funder’s choice. In our model, a researcher’s output increases in both capability and resources but is limited by the scarcer of the two, and the funder observes neither directly. This formulation lets us address questions raised by all four literatures: how widely funds should be spread (§7), what peer review is worth (§5), and when lotteries lose little (§7).

### 3. OUR MODEL

A funder faces a population of  $n$  researchers over  $T$  funding rounds. Researcher  $i$  has capability  $K_i$  and resources  $R_i$ . Initial capabilities and resources are drawn from Pareto distributions with tail parameters  $\alpha_K$  and  $\alpha_R$ ; smaller tail parameters produce more unequal fields, and a correlation parameter  $\rho$  lets capability and resources be correlated across researchers. The funder knows these distributions; it never observes any researcher’s capability or resources directly.

In each round, researcher  $i$  produces research output<sup>1</sup> at random, in discrete units, with expected output<sup>2</sup>

$$\lambda_i = A \Lambda(K_i, R_i),$$

where  $A$  is a productivity constant and  $\Lambda$  is a production function that increases in both arguments but is limited by the scarcer of the two. Our main analysis

<sup>1</sup> We are agnostic about the interpretation of output units: they may be interpreted as publications, as findings, or as units of scientific value.

<sup>2</sup> Output in each round is drawn from a Poisson distribution with rate  $\lambda_i$ , independently across researchers and rounds given the state.

takes  $\Lambda$  to be the harmonic mean:

$$\lambda_i = \frac{2AK_iR_i}{K_i + R_i}.$$

The harmonic mean makes capability and resources complements without making either an absolute requirement: a capable researcher with few resources still produces, by working more slowly or borrowing equipment, but no surplus of one input fully substitutes for a shortage of the other. Appendix C repeats our analysis for a family of production functions running from Cobb-Douglas, under which either input can substitute for the other, to Leontief, under which neither can.

Capabilities increase through research. Between rounds,

$$K_i \leftarrow K_i + \epsilon \Lambda(K_i, R_i),$$

where  $\epsilon$  is the compounding rate: doing research builds expertise, and the same bottleneck that limits output limits growth. Growth requires both inputs, and capability never declines. Resources, by contrast, do not accumulate: resources in a round are baseline plus grant,  $R_i = R_{i0} + g_i$ . The baseline  $R_{i0}$  recurs every round, representing the time and support a researcher has independent of the funder, while grants are consumed in the round they are given. Whatever lasts from a grant (skills, publications, standing) enters the model as capability.

The funder’s total budget is fixed before any outcomes are observed. We parameterize it by a budget scale  $b$ : at  $b = 0.5$ , the budget equals the community’s expected baseline resources for one round, and the budget scales linearly in  $b$  (Appendix D). The budget does not grow with the horizon: a longer horizon varies the timing of spending, never its amount. Strategies that do not plan ahead spend an equal share of the budget each round. Forward-looking strategies may spend on any schedule: each round, they re-plan the entire remaining budget over the remaining horizon and execute the current round’s allocation.

The funder learns as it goes. It observes each researcher’s track record, the output they have produced to date, as it accumulates, and two further signals are available. A resource signal, observed once at the start, gives a noisy estimate of each researcher’s baseline resources (noise  $\tau_R$ ), as one might obtain from institutional information. A review signal, drawn afresh in each round that a strategy uses it, gives a noisy estimate of capability (noise  $\tau_K$ ), as one might obtain from proposal evaluation and expert judgment. The review signal’s informativeness is governed by  $\tau_K$ : the smaller the noise, the more informative the review. Bayesian strategies update beliefs about every researcher after each round and allocate each dollar to the researcher whose expected output, given the funder’s current beliefs, rises most with it.

| Strategy                | Information                             | Planning          |
|-------------------------|-----------------------------------------|-------------------|
| No funding              | none                                    | none              |
| Uniform                 | none                                    | none              |
| Track record            | track record                            | none              |
| Myopic, without review  | track record + resource signal          | current round     |
| Forward, without review | track record + resource signal          | remaining horizon |
| Myopic, with review     | track record + resource signal + review | current round     |
| Forward, with review    | track record + resource signal + review | remaining horizon |

TABLE 1. The funding strategies compared.

We compare the strategies in Table 1. The Bayesian strategies differ along two dimensions: information, with or without the review signal, and planning, myopic (optimizing the current round’s expected output) or forward-looking (optimizing expected output over the whole remaining horizon). Three baselines complete the set: no funding, against which every result is normalized; uniform funding, which splits each round’s budget equally; and track-record funding, which allocates in proportion to the previous round’s output. Seed grants and lotteries, which modify these strategies, are introduced in §7.

Each of our conclusions rests on a proof or on simulations across the following parameter ranges; default values and the settings behind each figure are in Appendix D. The tail parameters  $\alpha_K$  and  $\alpha_R$  range from 1.3 to 3.5. Empirical distributions of research productivity have Pareto tail parameters near 1 to 2 (??); our range covers these values and extends to much more equal fields. The budget scale  $b$  ranges from 0.1 to 1, from a funder whose budget is small beside the community’s own resources to one able to relieve most researchers’ bottlenecks; §5 and §6 also examine larger budgets. The compounding rate  $\epsilon$  ranges from near zero, where funding raises only current output, to 0.85, where this round’s funding substantially raises later capability. The review noise  $\tau_K$  ranges from 0.05, nearly perfect review, to 20, nearly uninformative review, a range wide enough to contain the noise implied by empirical estimates of review’s predictive accuracy (§5). The correlation  $\rho$  ranges from  $-0.5$  to  $0.8$ , and the horizon  $T$  from one round to ten.

Randomness enters through the initial population draws, the signals, and realized output; capability growth follows expected output, so a researcher’s accumulation depends on their capability and resources rather than on luck in a given round. We score each strategy by expected output and compare strategies on identical simulated populations, and we report each strategy’s improvement over no funding in percent (Appendix D).

With the model in place, the funder’s problem is well defined: choose grants, given beliefs, to maximize expected total output over the horizon. Next we solve the case

in which the funder has complete information regarding researchers' capabilities and resources.

#### 4. THE OPTIMAL ALLOCATION: FUND THE CAPABILITY-RESOURCE GAP

First, consider the simple case where the funder has complete information regarding researchers' capabilities and resources. Further, we initially restrict our attention to a single round of funding, setting aside for now the growth of capability across rounds. The funder's problem is to divide the round's budget,  $B$ , among researchers so as to maximize expected output:

$$\max_{g_1, \dots, g_n \geq 0} \sum_i A \Lambda(K_i, R_{i0} + g_i) \quad \text{subject to} \quad \sum_i g_i \leq B.$$

Now, consider the marginal value of a dollar for such a funder. Under the harmonic production function,

$$\frac{\partial \lambda_i}{\partial g_i} = \frac{2A K_i^2}{(K_i + R_{i0} + g_i)^2}.$$

The marginal value of funding researcher  $i$  increases with their capability and decreases with the resources they already hold, including grants. An allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it, that is, when marginal values are equal across all funded researchers and no unfunded researcher's marginal value exceeds that common value.

**Proposition 1** (the gap rule). *For every researcher, the optimal grant fills the gap, if any, between the resources they hold and a target proportional to their capability:*

$$g_i^* = \max(c K_i - R_{i0}, 0),$$

where the constant  $c > 0$  is the value at which the grants sum to the budget.<sup>3</sup>

Each researcher's target is  $c$  times their capability, and their gap is the difference between that target and the resources they hold. The rule grants every researcher with a positive gap exactly their gap and grants nothing to the rest. When the budget is small,  $c$  is small: targets are low and few researchers have positive gaps. When the budget is large,  $c$  is large and funding extends across nearly the whole population (Corollary 1, Appendix A). In both cases the funded researchers are those with the least resources relative to their capability, and these are the researchers for whom a dollar has the greatest marginal value.<sup>4</sup> The condition

<sup>3</sup>  $c = \sqrt{2A/\nu} - 1$ , with  $\nu$  the common marginal value at the optimum. The proof is in Appendix A.

<sup>4</sup> The optimality condition behind the rule, funding to a common marginal value, is the water-filling solution familiar from information theory (?). If all researchers shared one capability, the

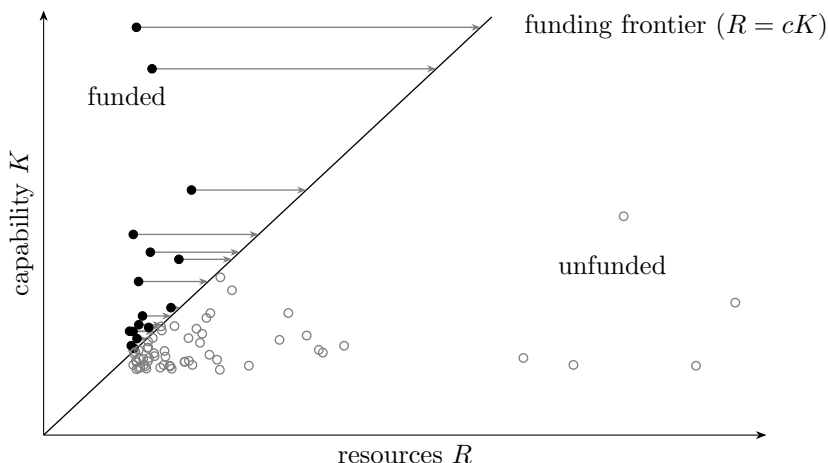


FIGURE 1. The gap rule in one population. Each point is a researcher, placed by resources and capability; the line is the funding frontier  $R = cK$ . Researchers below the frontier (filled) are funded, and each arrow is a grant, which moves its researcher exactly to the frontier. Researchers on or beyond the frontier (open) receive nothing.

$R_{i0} < cK_i$  draws a frontier through the population, the funding frontier, separating the funded from the unfunded (Figure 1).

The gap rule explains why each of the intuitive funding schemes discussed earlier fails. Funding the track record fails because output reflects resources as well as capability: a productive researcher whose resources approach or exceed their target has a small or negative gap, and an additional dollar adds little to their output. The failure is not merely a shortfall from the optimum: because track-record funding channels money toward researchers whose bottleneck is not resources, it can produce less output than spreading the budget evenly, that is, than not targeting at all. Funding the under-resourced fails for the opposite reason: scarcity is no evidence of capability, and dollars given to researchers of low capability add little output however scarce their resources. It, too, can produce less than not targeting at all.<sup>5</sup> Each scheme accounts for one of capability and resources, whereas the marginal value of a dollar is determined by how a researcher's resources compare with their capability, and so responds to both.

How much targeting matters depends on the budget. A researcher's expected output increases with every dollar of resources, but with diminishing marginal

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rule would reduce to filling each researcher's gap below one common target, the poverty-gap transfer of development economics (?); our rule scales the target with capability.

<sup>5</sup> Examples 1 and 2 in Appendix A make both failures exact with two researchers.

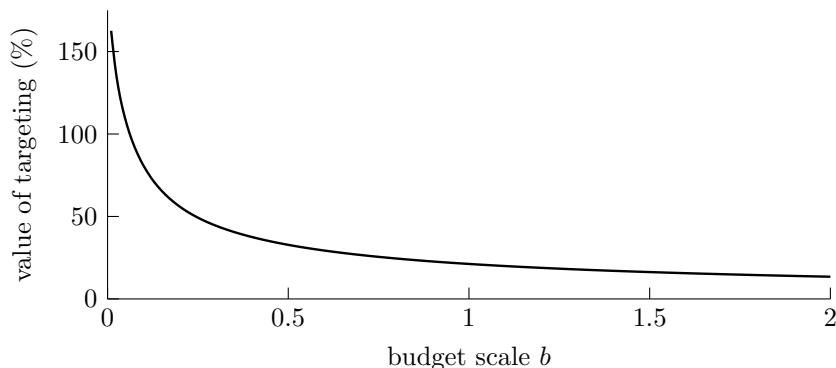


FIGURE 2. The value of targeting decreases in the budget. The additional expected output of the optimal allocation over uniform funding, as a fraction of uniform funding’s gain over no funding, against the budget scale  $b$ , for one population drawn from the default distributions (Corollary 2).

returns, and it never exceeds  $2AK_i$ : expected output is bounded above by a quantity set by capability, which it approaches but does not attain. When the budget is large, optimal and uniform funding therefore differ little in the output they produce: under either, every researcher’s output is near its upper bound, and the difference between the two vanishes as the budget increases (Corollary 2, Appendix A). When the budget is small, dollars are scarce, researchers differ in the marginal value of a dollar, and output depends on where each dollar goes. Targeting matters where the budget is small relative to researchers’ resources and little where the budget is large (Figure 2). This dependence on the budget recurs later: it sets how much output is lost when seed grants and lotteries override the optimal allocation, and it determines when optimal funding concentrates rather than spreads (both §7).

To use the gap rule, however, a funder requires an estimate of the gap between researchers’ capabilities and their resources. Next we turn to how such estimation can work: first from the track record alone, then with peer review added.

## 5. WHAT THE FUNDER CAN LEARN: TRACK RECORDS AND PEER REVIEW

**What the track record can and cannot supply.** The funder observes track records: each researcher’s realized output, round by round. A track record is evidence about the capability-resource gap, and the question is how far that evidence goes.

Consider first what output can identify in principle. Expected output is one number, produced by capability and resources jointly: every pair of capability and resources with  $2AKR/(K + R) = \lambda$  produces the same expected output  $\lambda$ . A

researcher of high capability and modest resources and a researcher of modest capability and abundant resources can therefore produce identical records. Output does carry some information about capability, a lower bound: since expected output never exceeds  $2AK$ , a researcher who produces  $\lambda$  has capability above  $\lambda/2A$ . Beyond that bound, the record cannot separate the two researchers above, and these are exactly the researchers the gap rule treats most differently: the first may have the largest gap in the population, the second may have none. Output, however carefully recorded, underdetermines what the funder most needs to know: the gap.

Our model measures how much output this underdetermination loses. A Bayesian funder relying on records alone allocates far from the gap rule, and further rounds of records improve its allocation only slowly; its output falls well short of what a funder with complete information would produce.<sup>6</sup> This shortfall is the most that any further information could recover; we ask next how much of it a review signal recovers.

The record is least informative exactly where the gap is largest. When a researcher's resources are small relative to their capability, expected output is approximately  $2AR$ : proportional to resources, nearly independent of capability (Figure 3). On a small grant, a researcher of capability 1 and a researcher of capability 100 produce nearly indistinguishable output. Small grants therefore yield almost no information about who is capable; separating capability from resources by observing output requires large grants, comparable to capability itself. §6 measures this and its consequences for timing.

What the record cannot supply, then, is a signal of capability separate from resources. Peer review can provide such a signal under the right conditions.

**Peer review: what a capability signal is worth.** Peer review enters our model as a noisy signal of each researcher's capability. The value of review is the additional expected output of a funder allocating with the review signal over the same funder allocating without it.

The direction of review's effect is what one would expect: a more informative signal brings the funder's allocation closer to the optimal allocation and its output closer to what complete information would deliver, and with a reliable enough signal the funder recovers most of the output that a funder relying on records alone

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<sup>6</sup> At the default parameters, the correlation between its grants and the optimal grants is 0.08 in the first round and reaches only 0.41 by round twenty; its output falls short of the complete-information output by 11.4 percent of no-funding output (24.6 percent under heavy-tailed capability).

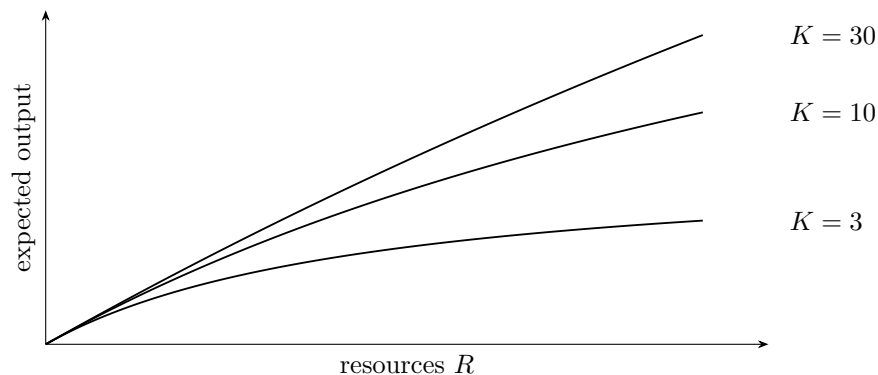


FIGURE 3. Small grants are uninformative about capability. Expected output against resources for researchers of capability  $K = 3, 10$ , and  $30$ . Where resources are small relative to capability the curves coincide, so researchers of very different capability produce nearly identical output; the curves separate only where resources are comparable to capability.

fails to obtain.<sup>7</sup> Two further effects are less obvious: (1) a funder relying on records alone does not catch up, and (2) what a review signal is worth varies widely with the field. The first effect has a structural cause. Capabilities compound across rounds while resources do not accumulate, so each researcher's output approaches the level their resources sustain, and output at that level carries less and less information about capability. Further rounds of records do improve the funder's allocation, but slowly: after twenty rounds of records the funder's allocation remains further from the optimal allocation than a funder with the review signal is after one (Figure 4).

Our main result is how much review is worth in different fields (Figure 5). Review is worth most where capability is heavy-tailed and the budget is tight, and in the heavy-tailed case even a signal of moderate informativeness captures most of its value. The reason is twofold. Where capability is heavy-tailed, most of the output a funder can add comes from getting resources to the few researchers of unusually high capability. And precisely those researchers are the easiest to identify: they stand far apart from everyone else, so even a noisy signal separates them. Identifying the most capable researchers captures most of the value and requires the least information. Making review more reliable beyond a modest level

<sup>7</sup> At the default parameters, the correlation between the funder's grants and the optimal grants increases from 0.22 at the noisiest signal in our range to 0.94 at the most reliable; under heavy-tailed capability ( $\alpha_K = 1.3$ ), from 0.34 to 0.96. The output shortfall relative to the complete-information output decreases from 10.4 to 1.6 percent of no-funding output; under heavy-tailed capability, from 17.2 to 1.7 percent.

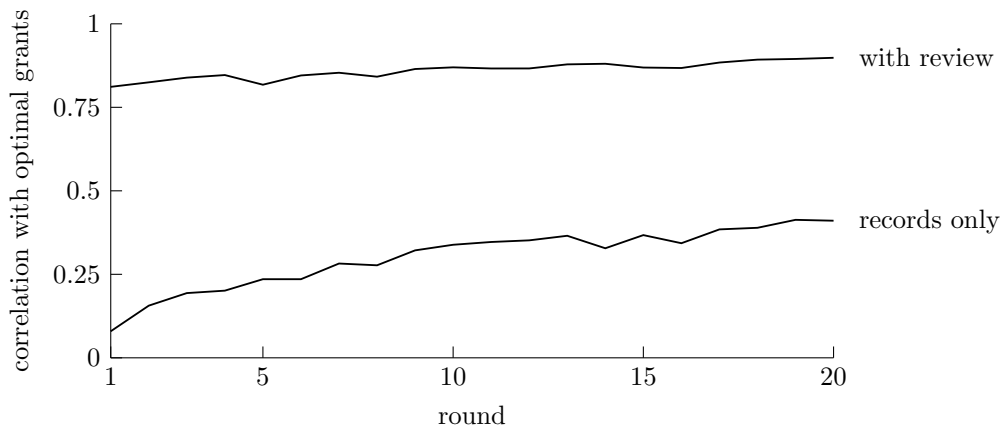


FIGURE 4. Records do not catch up. The correlation of each funder’s grants with the optimal grants, round by round. The records-only funder improves slowly and after twenty rounds remains further from the optimal allocation than the funder with the review signal is after one. Settings: Appendix D.

therefore adds little.<sup>8</sup> In the other fields, review is worth less. Where capability is spread more evenly, no researcher matters much more than another, so review of any informativeness adds little.<sup>9</sup> And where the budget is ample, optimal and uniform funding produce nearly the same output, so information about whom to fund has little left to add: review is worth most where the budget is small relative to researchers’ resources. This ordering of review’s value by field does not depend on our choice of production function (Appendix C).

A consequence concerns where review effort should go. In our model, what adds output is separating the few researchers of unusually high capability from the rest, and that separation requires little information; fine distinctions among the many applications of comparable merit add little output. A review system that spends most of its effort on those fine distinctions may be misallocating that effort.

These results bear on the dispute over peer review. Among funded NIH grants, percentile scores predict subsequent productivity barely better than chance; in our model, a signal of that accuracy is very noisy review.<sup>10</sup> Whether such review is worth having depends on the field: at the noisiest signal in our sweep, review still

<sup>8</sup> At three times the default signal noise, review retains 82 percent of its maximum value under heavy-tailed capability ( $\alpha_K = 1.3$ ), 44 percent at the default distribution ( $\alpha_K = 2$ ), and 10 percent where capability is spread more evenly ( $\alpha_K = 3.5$ ).

<sup>9</sup> At  $\alpha_K = 3.5$ , review’s value at its maximum is 2.5 percent of no-funding output, against 23 percent under heavy-tailed capability.

<sup>10</sup> An AUC of 0.54 (?) corresponds to  $\tau_K \gtrsim 20$  at the default parameters, the noisiest signal in our sweep. The mapping depends on the field: at the same noise, measured accuracy is higher under heavy-tailed capability, because the most capable researchers stand further apart.

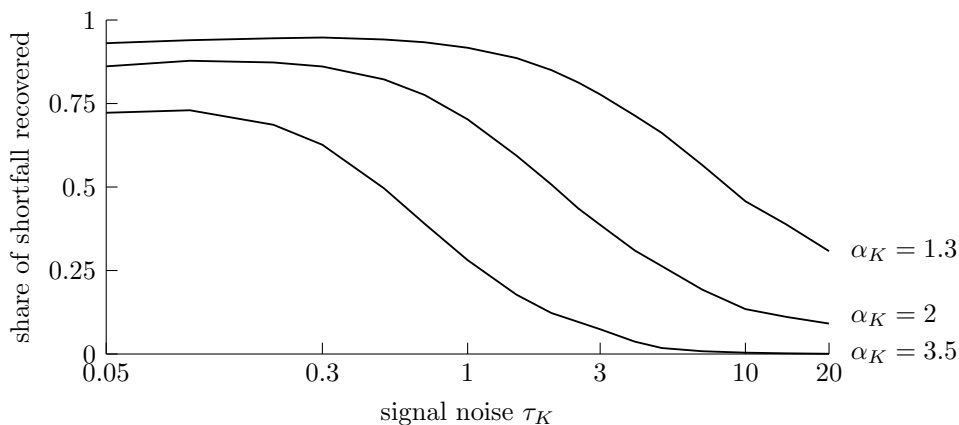


FIGURE 5. Review is worth most where capability is heavy-tailed. The share of the records-only shortfall that the review signal recovers, as review noise  $\tau_K$  increases, for three capability distributions. Settings: Appendix D.

recovers about a third of the records-only shortfall under heavy-tailed capability, under a tenth at the default distribution, and essentially none where capability is spread more evenly.<sup>11</sup> No single number settles the dispute; §8 returns to it.

One condition qualifies these results: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether: with moderately spread capability, overtrust turns review's value negative; with heavy-tailed capability it reduces the value without reversing it (Figure 6).<sup>12</sup> Overtrusting review loses more output than undertrusting it.

Review can inform whom to fund. Next we turn to when to fund: how funding should be spread across rounds, and how much that choice matters.

## 6. THE TIMING OF FUNDING

The funder holds a fixed budget for the whole horizon, and between rounds two things change: researchers produce output, which the funder observes, and capabilities compound. Up to now the funder has spent its budget in equal parts each round; this section asks whether it should spend early, evenly, or late, and how much the choice matters.

<sup>11</sup> 31, 9, and 0.1 percent of the records-only shortfall at  $\tau_K = 20$ , for  $\alpha_K = 1.3, 2$ , and  $3.5$ .

<sup>12</sup> A funder that overtrusts tenfold (true  $\tau_K = 3$ , belief 0.3) loses more than review's full value at the default capability distribution; a funder that undertrusts tenfold (true  $\tau_K = 0.3$ , belief 3) retains over half of it.

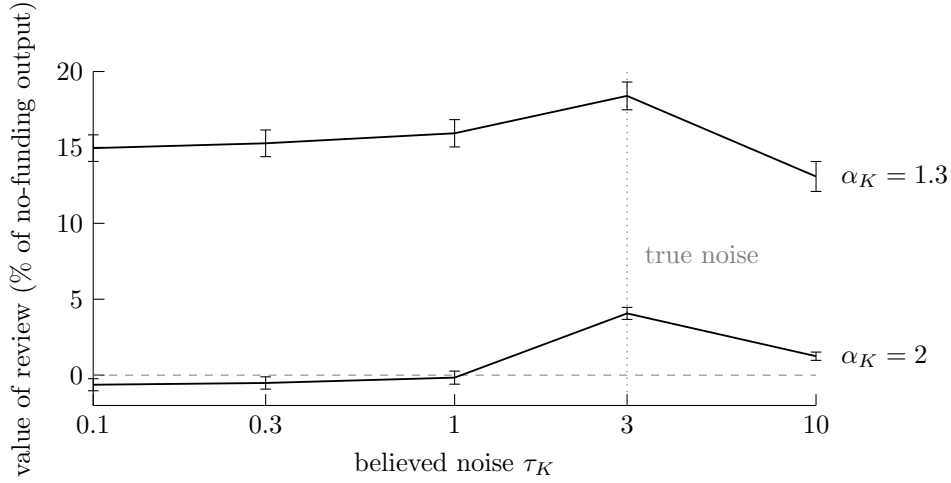


FIGURE 6. Overtrusting review loses more output than undertrusting it. The value of review as a percent of no-funding output, as the funder’s believed signal noise varies while the true noise is  $\tau_K = 3$  (dotted rule); bars are one standard error. At the default capability distribution, overtrust takes the value below zero. Settings: Appendix D.

The strongest argument for spending early concerns fields whose researchers have few resources. A researcher without resources produces nothing, so the funder observes nothing and no capability grows; money held for later rounds therefore produces neither evidence nor growth in the meantime. On this argument a funder facing such a field should spend at once. The argument fails. A funder that spends everything in the first round allocates before observing any output, and knows exactly as little as a funder that spends everything in the last round; in the extreme case of a field with no resources at all, the two schedules produce identical expected output. What the argument does establish is a complementarity: early spending enables observation and growth, and late spending exploits what early spending revealed. This complementarity favors putting some money early. It does not favor putting most of it early.<sup>13</sup>

Our simulations bear this out and say which should come first: observation, then money. Wherever capabilities compound at more than a negligible rate, the optimal schedule spends less than an even share in the first round and spends the bulk once output has revealed who is capable; spending late pays because money spent later is spent with more information, and because early output compounds

<sup>13</sup> Among schedules fixed in advance, the best schedule that spends more than an even share in the early rounds loses to the even schedule by 1.6 percent of the even schedule’s output at negligible compounding and by 3.5 percent at the default compounding rate (six rounds, community with few resources).

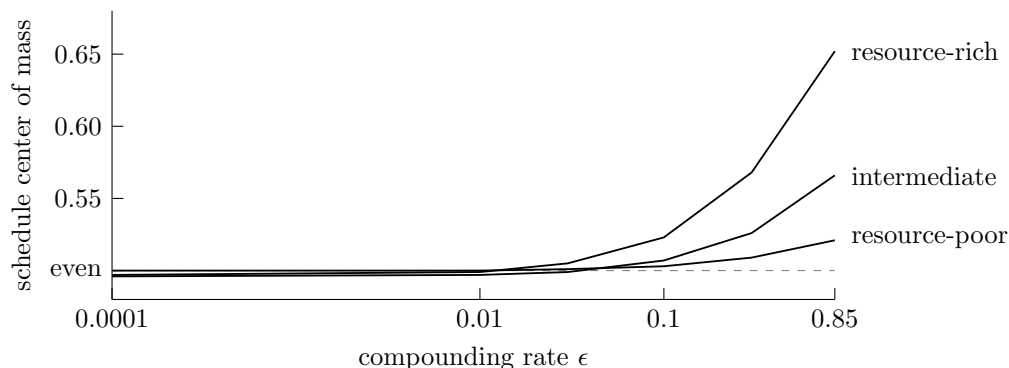


FIGURE 7. The compounding rate, not the scarcity of baseline resources, sets how far spending shifts toward later rounds. The schedule’s center of mass (0.5 is even; higher is later) as the compounding rate  $\epsilon$  increases, for three baseline resource scales. Settings: Appendix D.

on its own (Figure 7). Scarce baseline resources raise the early share and decreases the intensity of the effect, but does not reverse it.<sup>14</sup>

How much a funder can learn from its own grants depends on their size. A researcher on a small grant produces output nearly proportional to the grant and nearly independent of capability, so a funder whose grants are small learns almost nothing it can exploit later: in a field whose researchers have few resources, with no review signal, such a funder gains almost nothing over uniform funding. Large grants change this. As grants approach the scale of capability itself, output separates the capable from the rest, and the funder recovers most of the value of discrimination, on a schedule that spends a small share early, so that output can be observed, and the rest once the funder has learned who is capable (Figure 8).<sup>15</sup>

The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded: that output compounds and reveals capability while the funder waits. Growth from grant-added output alone would shift the schedule slightly toward earlier rounds, since earlier grants have longer to compound.<sup>16</sup>

<sup>14</sup> Every parameter setting whose optimal schedule spends more than an even share early has a compounding rate  $\epsilon \leq 0.03$ , and there the schedule’s center of mass is within 0.005 of even.

<sup>15</sup> In a community whose researchers have few resources, with heavy-tailed capability and no review signal, over six rounds, the informed funder’s gain over uniform funding increases from a third of one percent of uniform funding’s output at the default grant size to nine percent at six times that size and ten percent at twelve times, roughly the gain a reliable review signal delivers at the default size.

<sup>16</sup> With growth from unfunded output switched off, the schedule’s center of mass is 0.48 to 0.49, below even; with growth from grant-added output switched off, 0.53 to 0.68 (five rounds).

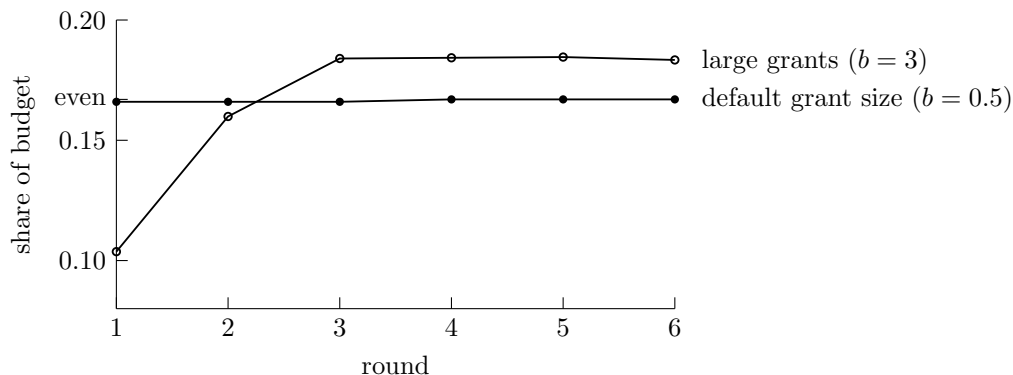


FIGURE 8. With large grants the funder spends a small share first and the rest once it has observed output. The share of the budget spent in each of six rounds, in a field whose researchers have few resources, with heavy-tailed capability and no review signal, at the default grant size (filled) and at six times that size (open). Settings: Appendix D.

Finally, the choice of schedule is worth less than the signal provided by grant peer review, in every setting we test. Across all horizons and compounding rates in our sweeps, the gain from choosing the schedule, over spending equal installments each round, stays below the review signal’s value: at the default compounding rate and below it is smaller by an order of magnitude or more, and it approaches the signal’s value only at the joint extreme of the highest compounding rate and the longest horizon we consider (Figure 9).<sup>17</sup> The gain increases roughly with the square of the horizon over short horizons and levels off beyond five rounds. And it exists only where capability and resources are complements, as in our production function: under Cobb-Douglas production the value of scheduling is indistinguishable from zero and the optimal schedule stays within about a hundredth of even (Appendix C, Table 2). For a funder thinking about how to maximize their impact, choosing whom to fund outweighs choosing when.

Next we turn to two alternatives to targeting that recur in funding policy: uniform seed grants and lotteries.

## 7. SPREADING FUNDS: SEED GRANTS, LOTTERIES, AND CONCENTRATION

Targeting sends each dollar to the researcher expected to produce the most with it. Two alternatives to targeting recur in funding policy, and each has a rationale. Uniform seed grants give every researcher the same amount, so that a researcher whom review misjudges still receives something; in practice a funder gives out part

<sup>17</sup> As a fraction of no-funding output, the scheduling gain ranges from indistinguishable from zero to 0.7 percent across the thirty-two tested parameter settings; the review signal’s value in the same settings ranges from 0.8 to 7.5 percent.

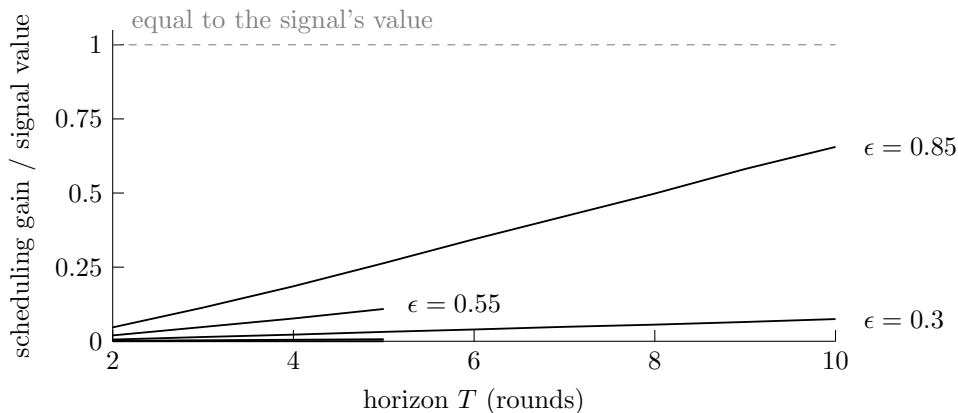


FIGURE 9. Choosing whom outweighs choosing when. The gain from choosing the schedule, as a share of the review signal’s value, by horizon and compounding rate  $\epsilon$ . In every setting the gain is below the signal’s value, and it approaches that value only at the highest rate and the longest horizon. Settings: Appendix D.

of its budget as seed grants and targets the rest. Lotteries give grants to applicants chosen at random, usually from among those who pass a review threshold; their advocates argue that review cannot reliably rank the applicants it deems fundable, that review is costly, and that review is biased. Both alternatives give up part of targeting on purpose. Our model can say how much expected output is lost by giving up targeting, field by field, and where the loss is small. What the alternatives save, in the cost of review and in the effort researchers spend on proposals, lies outside our model and has been studied by others (§2).

Both alternatives lose the same thing: the value of targeting, the additional expected output that the optimal allocation produces over an equal division of the same budget. How much output is lost by giving up targeting depends on the field. For a funder allocating with records and review: in a heavy-tailed field with reliable review and a tight budget, targeting adds more output than uniform funding’s entire gain over no funding; in the default field at the largest budget we sweep, targeting adds about a quarter of that gain.<sup>18</sup> Giving up targeting loses little output under the same conditions that make review worth little: where capability is evenly spread, the budget is large, or review is uninformative.

Consider uniform seed grants first. The output lost increases faster than the fraction of the budget given out as seed grants: giving out half of the budget as seed

<sup>18</sup> The review-informed funder’s gain over uniform funding, as a fraction of uniform funding’s gain over no funding, decreases from 1.17 to 0.72 across budget scales  $b = 0.1$  to 1 with heavy-tailed capability and reliable review ( $\tau_K = 0.3$ ), and from 0.74 to 0.25 in the default field with the default signal.

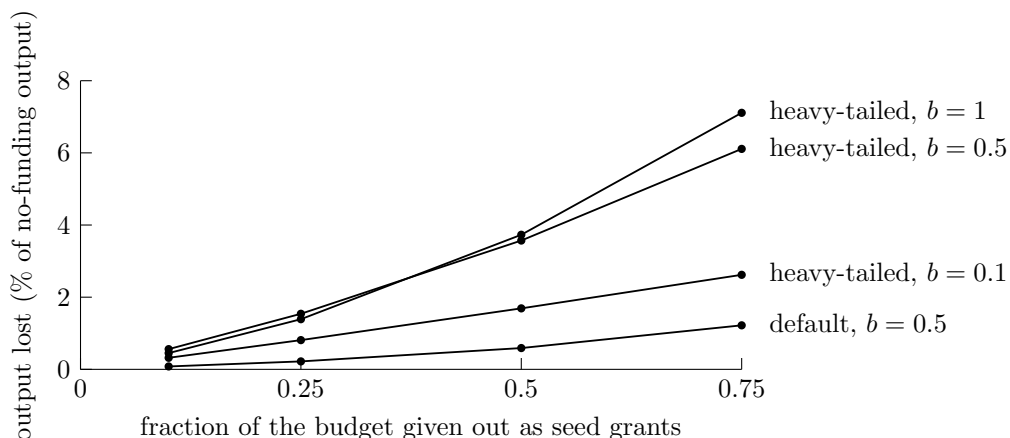


FIGURE 10. The output lost to uniform seed grants. Expected output lost, as a percent of no-funding output, when a review-informed funder gives out a fraction of its budget as equal grants to all researchers and targets the rest. The loss grows faster than the fraction, and its size is set by what targeting is worth in the field. Settings: Appendix D.

grants loses more than twice the output that giving out a quarter loses (Figure 10). The size of the loss is set by what targeting is worth in the field. When a review-informed funder gives out three quarters of its budget as seed grants, expected output falls by about six percent of no-funding output in a heavy-tailed field with reliable review, and by about one percent in the default field. When the whole budget is given out as seed grants, the scheme is uniform funding, and the full value of targeting is lost. Whether the reasons for seed grants justify the loss is a judgment our model does not make; the model supplies the size of the loss.

Now consider lotteries. A lottery gives each applicant in its pool a chance at a full grant instead of a certain smaller one, and on average the applicant receives the same amount either way. Because expected output increases with resources at a diminishing rate, a certain grant produces more expected output than a chance at a larger grant of the same average size. Dividing the pool's money equally among the pool therefore produces more expected output than any lottery over the same pool, and a lottery over all applicants produces less than uniform funding (Proposition 2, Appendix B). To measure how much less, and how lotteries compare with selection by review, we evaluate five schemes in a single round (Figure 11): equal division among the fifth of researchers with the highest review scores (selection by review); equal division among the top two fifths; the partial lottery that some funders run, which funds half of the top two fifths at random with full-size grants; uniform funding; and a lottery over all researchers with full-size grants. In a heavy-tailed field with a tight budget, selection by review gains

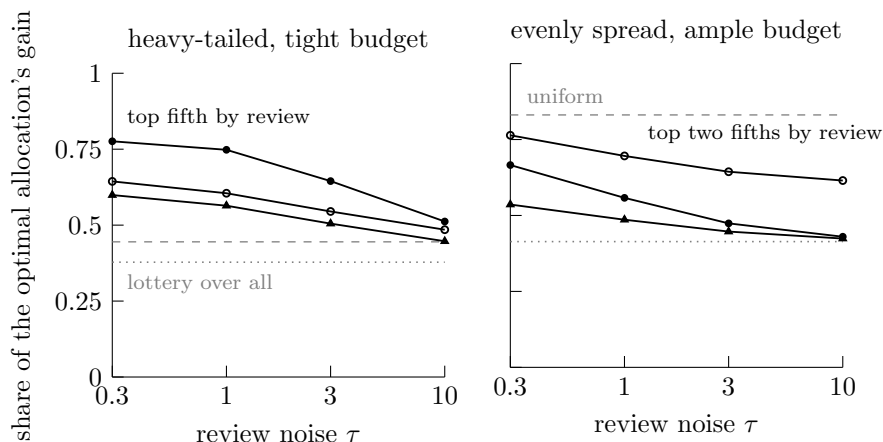


FIGURE 11. What five funding schemes gain, by field. Each scheme's gain over no funding as a share of the optimal allocation's gain, as review noise increases, in a heavy-tailed field with a tight budget (left) and an evenly spread field with a large budget (right). Filled circles: equal division among the top fifth by review score; open circles: among the top two fifths; triangles: the partial lottery; dashed: uniform funding; dotted: a lottery over all researchers. Settings: Appendix D.

about three quarters of what the optimal allocation gains when review is reliable and about half when review is very noisy. The partial lottery gains less, and most of the difference comes from ignoring the review ranking within the top two fifths rather than from the randomness itself.<sup>19</sup> The lottery over all researchers gains least, less than uniform funding. In an evenly spread field with a large budget, the ordering reverses: uniform funding gains more than four fifths of what the optimal allocation gains, and every scheme that concentrates grants, whether by review score or by chance, gains less.

A lottery among applicants above a review threshold therefore loses little output only where the review ranking it ignores is worth little: replacing that ranking by chance loses a few hundredths of the optimal allocation's gain in an evenly spread field or when review is very noisy, and about a sixth of that gain in a heavy-tailed field with reliable review.<sup>20</sup> The lottery debate rarely asks whether to concentrate grants at all. In an evenly spread field, our model's answer is to spread the budget in small grants across many researchers; a lottery among a few

<sup>19</sup> At review noise  $\tau = 1$ : selection by review gains 0.75 of what the optimal allocation gains, the equal division among the top two fifths 0.61, and the partial lottery 0.56; of the partial lottery's 0.18 shortfall, ignoring the ranking accounts for 0.14 and the randomness for 0.04.

<sup>20</sup> The partial lottery's shortfall relative to selection by review: 0.01 to 0.07 of the optimal allocation's gain in the evenly spread field and at the noisiest review in the heavy-tailed field; 0.14 to 0.18 in the heavy-tailed field with reliable to moderately noisy review.

applicants above a review threshold produces less. Whether a lottery’s savings in review costs and proposal effort outweigh the output it loses depends on both quantities; the contest models cited in §2 measure the savings, and our model measures the loss.

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. One might hope that the production function settles this question: the harder it is to substitute resources for capability, the more the money should go to the few researchers who can use it. To test this, we embed our production function in a family indexed by an exponent  $\gamma$ , from Cobb-Douglas ( $\gamma = 0$ ), under which a shortfall in either input can be offset by more of the other, through our form ( $\gamma = -1$ ), to Leontief, under which output depends only on the scarcer input; the harder it is to substitute one input for the other, the stronger the complementarity (Appendix C). The production function does not settle the question (Figure 12). With a tight budget in a heavy-tailed field, stronger complementarity does concentrate the review-informed funder’s grants. With an ample budget, stronger complementarity spreads them, a pattern consistent with a funder that, once the budget allows, brings every researcher’s resources up toward the level their capability can use. Where capability is evenly spread and the budget ample, the relation is not even monotone: concentration peaks at an intermediate complementarity and declines toward both ends.<sup>21</sup> How widely funds should be spread is set jointly by the production function, the budget, and the field’s capability distribution; no one of the three decides it alone.

Next we take stock: what the model implies for funders, and what its assumptions leave out.

## 8. DISCUSSION

We have compared the expected impact on research output of targeted funding to what it would be under equal division of the same budget, and to what it would be in the absence of any contribution by the funder. Our results isolate one quantity as key to strategies for optimal grant funding: the value of targeting researchers with the greatest gap between their capabilities and their resources. Two features of an area of research determine this value. The first is the funder’s budget relative to researchers’ own resources. Where the budget is small, dollars are scarce and

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<sup>21</sup> Gini coefficients of the review-informed funder’s grants, from Cobb-Douglas to Leontief: 0.92 to 0.95 in a heavy-tailed field with a tight budget; 0.35 to 0.21 in the default field with an ample budget; 0.63 to 0.45 in a heavy-tailed field with an ample budget. In the evenly spread field with an ample budget the Gini coefficient is 0.16 at Cobb-Douglas, peaks at 0.196 near  $\gamma = -6$ , and falls to 0.185 at  $\gamma = -12$  (3.5 standard errors above the  $\gamma = -12$  level).

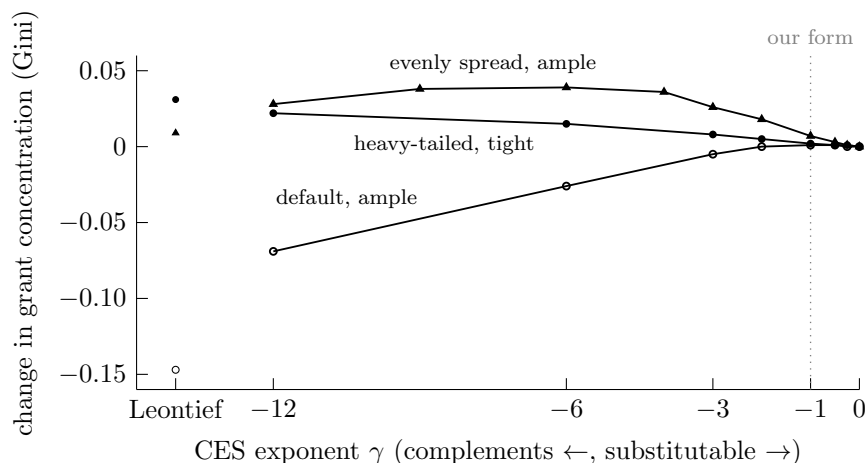


FIGURE 12. Complementarity concentrates funding only when the budget is small. The change in the Gini coefficient of the review-informed funder’s grants, relative to its Cobb-Douglas level, as the production function moves from Cobb-Douglas ( $\gamma = 0$ ) toward Leontief (detached marks at the left edge), for three field-and-budget combinations. Settings: Appendix D.

where they go matters; where the budget can relieve most researchers’ bottlenecks, the optimal allocation and equal division produce nearly the same output (§4, Figure 2). The second is how unequally capability is distributed. Where a few researchers can produce far more than the rest, getting resources to those few is most of what a funder can add (§5).

We showed how much various sources of information contribute to the effectiveness of targeting this gap. Track records are modestly predictive of future output, but they underdetermine whether a researcher’s productivity is bottlenecked by resources or by capability, and further rounds of records improve the funder’s allocation only slowly (§5). Peer review supplies what records cannot, a signal of capability itself, and the value of that signal increases with the two features that set the value of targeting and with the signal’s informativeness; where capability is heavy-tailed, even noisy review recovers most of that value (§5). Large grants make records informative, and spending later lets the funder allocate with more information, but the gain from choosing when to fund is smaller than the gain from the review signal at every horizon and compounding rate we sweep (§6). We also measured what a funder loses by forgoing targeting: seed grants and lotteries lose most output where the value of targeting is largest, a lottery over any pool produces less expected output than an equal division of the same money over that pool, and whether funding should be concentrated or spread depends on the budget and the field, not on the production function alone (§7).

These results have a practical consequence for funders. No funding mechanism is good or bad in itself. Peer review, seed grants, and lotteries each perform better in some fields than in others, and the two features of the field decide which. A funder choosing among mechanisms should therefore begin by estimating the two features for its own field. The funder can compute the first feature, its budget relative to researchers' total resources, from its own budget and what it knows about the field. The inequality of capability is harder to observe. The natural proxy for the distribution of capability is the distribution of output, but where grants are small, output tracks resources rather than capability, so the output distribution understates capability inequality (§5). Two cases show what follows. A small foundation funding a heavy-tailed field faces the largest value of targeting per dollar. For this foundation, review directed at identifying the few researchers of unusually high capability is worth most, and even noisy review recovers most of that value; seed grants lose the most output; and a lottery among applicants above a review threshold loses about a sixth of what selection by review gains. A large agency funding an evenly spread field faces a small value of targeting. For this agency, review of any informativeness adds little; seed grants and lotteries lose little; and the best use of the budget is small grants to many researchers.

Three further implications concern peer review. The first concerns where review effort should go. Since most of review's value comes from separating the few researchers of unusually high capability from the rest, and fine distinctions among the many applications of comparable merit add little output, a review system that spends most of its effort ranking the middle of the distribution spends that effort where the model locates the least output (§5). The second concerns how to read the evidence on review's efficacy. The finding that percentile scores barely predict productivity comes from funded NIH grants with scores in the top fifth (?), and the studies that find modest predictive power also use funded grants (?). Both kinds of study measure prediction among applicants that review has already selected, that is, after review has made the separation the model values most, and within the range of applicants where the model expects review to add least. A weak association within that band is consistent with review of considerable value over the whole applicant pool. For the same reason, the mapping in §5 from measured accuracy to signal noise, which treated the measured accuracy as review's accuracy over the whole pool, understates review's informativeness. The third concerns how much a funder should trust review. A funder that treats review as more reliable than it is loses more output than a funder that treats review as noisier than it is (§5, Figure 6). A funder uncertain about review's accuracy should therefore err toward undertrust.

The model holds several things fixed, and its results hold within those limits. There is one funder. Competition among funders, and the process by which past funding attracts future funding across funders, lie outside the model, though the gap rule itself directs grants away from researchers whose resources are already large. Researchers do not respond strategically. Proposal effort, risk choice, and the cost of review lie outside the model; the contest models cited in §2 cover that ground. Output is a single number per round. Projects do not differ in kind, and exploring new directions has no value in the model, so results about funding unexplored regions of a field have no counterpart here. The review signal is noisy but unbiased. Bias and conservatism in review, which are among the strongest arguments for lotteries, are not modeled. Researchers do not enter or leave the field, and have no careers. Grants are consumed and inputs are complements; the timing results depend on complementarity and the information results do not (Appendix C). The funder is an idealization that knows the model and updates correctly, so the values of information reported here are what such a funder extracts, not what a real funder would extract. Finally, the numbers we report are comparisons across the parameter values we sweep, and none is calibrated to a real funder. What applies to a real field is the qualitative features of the dynamics that organize the results.

The model makes three predictions that data on funded research can test. First, measures that stand in for capability, such as prior record or review score, should predict output better the larger a grant is relative to the recipient’s other resources, because on small grants output is approximately proportional to resources and nearly independent of capability (§5, Figure 3). A test compares the predictive power of review scores across grant sizes within one applicant pool. Second, how well review scores predict output, measured over the whole applicant pool rather than among funded grants, should be higher in fields whose output distributions have heavier tails (§5). A test compares fields, or one funder’s programs, using the full range of scores where the outcomes of funded and unfunded applicants can both be observed. Third, where funders run lotteries beside review, the output lost to the lottery should be larger in heavier-tailed fields (§7, Figure 11). The lottery programs now running at several funders supply the comparison. The three tests use data that funders already hold: grant sizes and recipients’ other funding, review scores and outcomes over whole applicant pools, and the outputs of lottery-funded and review-funded grants where a funder runs both.

## 9. CONCLUSION

A funder with a fixed budget and imperfect knowledge of researchers faces one problem, and our model gives it one answer. The optimal allocation funds the

gap between each researcher’s capability and their resources, and neither of the intuitive rules, funding the track record or funding the under-resourced, does so, because each tracks only one of the two. What the funder can learn about the gap sets everything else. Records alone underdetermine it. Peer review can supply the missing signal of capability, and what review is worth, what seed grants and lotteries lose, and whether to concentrate or spread all turn on the same two features of a field: how much of researchers’ resources the funder’s budget can supply, and how unequal capability is. Where the budget is tight and capability heavy-tailed, targeting is worth a great deal, even noisy review captures most of that value, and giving up targeting loses much of it. Where the budget is ample and capability evenly spread, little output is at stake and spreading funds widely loses almost nothing. In every field we examine, choosing whom to fund matters more than choosing when. A funder deciding among review, lotteries, and seed grants should therefore begin by asking which kind of field it is funding.

## Appendices

### A. DERIVATION OF THE GAP RULE

**The within-round allocation problem.** A population of  $n$  researchers. Researcher  $i$  has capability  $K_i$  and baseline resources  $R_{i0}$ , both fixed within the round. A grant  $g_i \geq 0$  adds to researcher  $i$ ’s resources for the round, so that their expected output is  $A \Lambda(K_i, R_{i0} + g_i)$  with  $\Lambda(K, R) = 2KR/(K + R)$ , the production function of §3, and  $A > 0$  the common productivity scale. Write

$$\varphi_i(g) = \frac{2A K_i (R_{i0} + g)}{K_i + R_{i0} + g}$$

for researcher  $i$ ’s expected output as a function of their grant alone.

Throughout,  $A > 0$ ,  $K_i > 0$  and  $R_{i0} \geq 0$  for every  $i$ , and the round’s budget is  $B > 0$ . Nothing else about the model is used, so the result holds for every realized population and every round’s budget; across rounds, the constant  $c$  and the funded set are recomputed from the new population state.

The funder observes  $(K_i, R_{i0})_{i=1}^n$  and divides the budget so as to maximize expected output:

$$(P) \quad \max_{g_1, \dots, g_n \geq 0} \sum_{i=1}^n \varphi_i(g_i) \quad \text{subject to} \quad \sum_{i=1}^n g_i \leq B.$$

**Three lemmas.**

**Lemma 1** (the return to a grant). *For each  $i$ , on  $g \in [0, \infty)$ : (i)  $\varphi_i$  is strictly increasing, with  $\varphi'_i(g) = 2AK_i^2/(K_i + R_{i0} + g)^2 > 0$ ; (ii)  $\varphi_i$  is strictly concave:  $\varphi''_i(g) = -4AK_i^2/(K_i + R_{i0} + g)^3 < 0$ ; (iii)  $\varphi'_i(g) < 2A$  whenever  $R_{i0} + g > 0$ , and  $\varphi_i(g) \uparrow 2AK_i$  as  $g \rightarrow \infty$ .*

*Proof.* Write  $S = K_i + R_{i0} + g$ . Then  $\varphi_i(g) = 2AK_i(S - K_i)/S = 2AK_i - 2AK_i^2/S$ , and (i) and (ii) follow by differentiating in  $g$  (note  $dS/dg = 1$ ). For (iii):  $\varphi'_i(g) = 2AK_i^2/S^2 < 2A$  exactly when  $S > K_i$ , that is, when  $R_{i0} + g > 0$ ; and  $\varphi_i(g) = 2AK_i - 2AK_i^2/S \rightarrow 2AK_i$  as  $S \rightarrow \infty$ .  $\square$

Lemma 1 says that the marginal value of a dollar to researcher  $i$  increases with their capability and decreases with the resources they already hold, that each dollar granted lowers the value of the next, and that expected output never exceeds  $2AK_i$ .

**Lemma 2** (existence, uniqueness, exhaustion). *Problem (P) has exactly one solution  $g^* = (g_1^*, \dots, g_n^*)$ , and it exhausts the budget:  $\sum_i g_i^* = B$ .*

*Proof.* The feasible set is compact and convex and the objective  $f(g) = \sum_i \varphi_i(g_i)$  is continuous and, by lemma 1(ii), strictly concave, so a maximizer exists and is unique. If  $\sum_i g_i^* < B$ , increasing any one  $g_i^*$  slightly stays feasible and raises  $f$  (lemma 1(i)), contradicting optimality.  $\square$

**Lemma 3** (equal marginal value). *A feasible allocation  $g$  with  $\sum_i g_i = B$  solves (P) if and only if there is a  $\nu > 0$  such that*

$$\varphi'_i(g_i) = \nu \text{ for every } i \text{ with } g_i > 0, \quad \varphi'_i(0) \leq \nu \text{ for every } i \text{ with } g_i = 0.$$

*Proof. Necessity.* Let  $g^*$  solve (P). Since  $B > 0$  is exhausted (lemma 2), some researcher is funded:  $g_i^* > 0$  for some  $i$ . Fix any such  $i$  and any  $j \neq i$ , and for  $0 < \delta \leq g_i^*$  consider the transfer that replaces  $g_i^*$  by  $g_i^* - \delta$  and  $g_j^*$  by  $g_j^* + \delta$ , leaving the rest unchanged. The result is feasible, so optimality gives

$$\varphi_j(g_j^* + \delta) - \varphi_j(g_j^*) \leq \varphi_i(g_i^*) - \varphi_i(g_i^* - \delta).$$

Dividing by  $\delta$  and letting  $\delta \downarrow 0$  yields  $\varphi'_j(g_j^*) \leq \varphi'_i(g_i^*)$ . If  $j$  is also funded, the same argument with  $i$  and  $j$  exchanged gives the reverse inequality, so all funded researchers share a common marginal value; call it  $\nu = \varphi'_i(g_i^*) > 0$  (positivity by lemma 1(i)). For unfunded  $j$ , the displayed inequality reads  $\varphi'_j(0) \leq \nu$ .

*Sufficiency.* Let  $g$  satisfy the condition with multiplier  $\nu$ , and let  $h$  be any feasible allocation. Concavity of  $\varphi_i$  gives, for every  $i$ ,  $\varphi_i(h_i) \leq \varphi_i(g_i) + \varphi'_i(g_i)(h_i - g_i)$ . For

funded  $i$ ,  $\varphi'_i(g_i) = \nu$ , so  $\varphi'_i(g_i)(h_i - g_i) = \nu(h_i - g_i)$ . For unfunded  $i$ ,  $g_i = 0$  and  $h_i - g_i = h_i \geq 0$ , so  $\varphi'_i(0)(h_i - g_i) \leq \nu(h_i - g_i)$ . Summing over  $i$ :

$$f(h) \leq f(g) + \nu \sum_i (h_i - g_i) = f(g) + \nu \left( \sum_i h_i - B \right) \leq f(g),$$

since  $h$  is feasible and  $\nu > 0$ . So  $g$  solves (P).  $\square$

Lemma 3 is the formal content of the sentence in § 4: an allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it.

**The proposition.**

**Proposition 1** (the gap rule). *The unique solution of (P) is*

$$g_i^* = \max(c K_i - R_{i0}, 0), \quad i = 1, \dots, n,$$

where  $c > 0$  is the unique constant satisfying the budget identity  $\sum_{i=1}^n \max(c K_i - R_{i0}, 0) = B$ . Equivalently,  $c = \sqrt{2A/\nu} - 1$ , where  $\nu \in (0, 2A)$  is the common marginal value of lemma 3.

*Proof.* Let  $g^*$  be the unique solution (lemma 2) and  $\nu$  the multiplier that lemma 3 associates with it. The proof proceeds in four steps.

*Step 1:*  $\nu \in (0, 2A)$ , so  $c = \sqrt{2A/\nu} - 1$  is well defined and positive. Positivity of  $\nu$  is part of lemma 3. For the upper bound: some researcher  $i$  is funded, and for them  $R_{i0} + g_i^* \geq g_i^* > 0$ , so lemma 1(iii) gives  $\nu = \varphi'_i(g_i^*) < 2A$ . Then  $2A/\nu > 1$ , so  $c = \sqrt{2A/\nu} - 1 > 0$ .

*Step 2:* funded researchers receive exactly their gap. Let  $g_i^* > 0$ . By lemma 3,  $2A K_i^2 / (K_i + R_{i0} + g_i^*)^2 = \nu$ . Solving, and taking the positive square root (both sides of the next display are positive),

$$K_i + R_{i0} + g_i^* = K_i \sqrt{2A/\nu} = (1 + c) K_i,$$

so  $g_i^* = c K_i - R_{i0}$ . Since  $g_i^* > 0$ , this equals  $\max(c K_i - R_{i0}, 0)$ .

*Step 3:* unfunded researchers have no gap. Let  $g_i^* = 0$ . By lemma 3,  $2A K_i^2 / (K_i + R_{i0})^2 = \varphi'_i(0) \leq \nu$ , so  $K_i + R_{i0} \geq K_i \sqrt{2A/\nu} = (1 + c) K_i$ , that is,  $R_{i0} \geq c K_i$ . Then  $\max(c K_i - R_{i0}, 0) = 0 = g_i^*$ .

Steps 2 and 3 establish  $g_i^* = \max(c K_i - R_{i0}, 0)$  for every  $i$ , and the budget identity is exhaustion (lemma 2). It remains to show  $c$  is the only constant satisfying that identity.

*Step 4: the budget identity pins  $c$  uniquely.* Define  $G(c) = \sum_{i=1}^n \max(c K_i - R_{i0}, 0)$  for  $c \geq 0$ .  $G$  is continuous and nondecreasing, with  $G(0) = 0$  and  $G(c) \rightarrow \infty$  as  $c \rightarrow \infty$ . Moreover  $G$  is strictly increasing wherever it is positive: if  $G(c) > 0$ , then  $c K_j > R_{j0}$  for some  $j$ , and for any  $c' > c$ ,  $G(c') \geq G(c) + K_j (c' - c) > G(c)$ . Now suppose  $c_1 < c_2$  both satisfy  $G(c) = B$ . Since  $B > 0$ ,  $G(c_1) > 0$ , so  $G(c_2) > G(c_1) = B$ , a contradiction.  $\square$

### Consequences.

**Corollary 1** (funding frontier and budget comparative statics). *(i) Researcher  $i$  is funded,  $g_i^* > 0$ , exactly when  $K_i > R_{i0}/c$ . (ii) As a function of the budget  $B$ , the constant  $c(B)$  is continuous and strictly increasing; hence the funded set is nondecreasing in  $B$ , every funded researcher's grant  $c(B) K_i - R_{i0}$  is strictly increasing in  $B$ , and the common marginal value  $\nu(B) = 2A/(1+c(B))^2$  is strictly decreasing in  $B$ . Both parts follow from the formula and Step 4.*

**Corollary 2** (the value of targeting vanishes in the budget). *Let  $f^*(B)$  denote the optimal value of (P) and  $f^u(B) = \sum_i \varphi_i(B/n)$  the value of uniform funding. Then  $0 \leq f^*(B) - f^u(B)$  for every  $B$ , and  $f^*(B) - f^u(B) \rightarrow 0$  as  $B \rightarrow \infty$ .*

*Proof.* The first inequality holds because uniform funding is feasible. For the second: by lemma 1(iii),  $\varphi_i(g) < 2AK_i$  for every  $g$ , so  $f^*(B) \leq 2A \sum_i K_i$ ; and  $f^u(B) = \sum_i \varphi_i(B/n) \rightarrow 2A \sum_i K_i$  as  $B \rightarrow \infty$ . Hence  $0 \leq f^*(B) - f^u(B) \leq 2A \sum_i K_i - f^u(B) \rightarrow 0$ .  $\square$

Since uniform funding's own gain over no funding does not vanish (it approaches  $2A \sum_i K_i - \sum_i \varphi_i(0) > 0$ ), the same conclusion holds for the value of targeting measured relative to that gain, the quantity plotted in Figure 2.

**Example 1** (funding the track record can produce less than uniform funding). *Two researchers,  $A = 1$ , budget  $B = 2$ . Researcher 1:  $K_1 = 8$ ,  $R_{10} = 8$  (productive and well resourced; expected output  $\varphi_1(0) = 8$ ). Researcher 2:  $K_2 = 4$ ,  $R_{20} = 1$  (capable but with few resources;  $\varphi_2(0) = 8/5$ ). Allocating in proportion to expected output gives  $g = (5/3, 1/3)$  and total output  $570/53 \approx 10.75$ . Uniform funding,  $g = (1, 1)$ , gives  $568/51 \approx 11.14$ . The gap rule ( $c = 3/4$ ) gives  $g^* = (0, 2)$  and  $80/7 \approx 11.43$ .*

**Example 2** (funding the under-resourced can produce less than uniform funding). *Two researchers,  $A = 1$ , budget  $B = 2$ . Researcher 1:  $K_1 = 10$ ,  $R_{10} = 5$ . Researcher 2:  $K_2 = 1/2$ ,  $R_{20} = 0$  (the scarcest resources in the population). Directing the budget to the researcher with the fewest resources gives total output  $112/15 \approx 7.47$ . Uniform funding gives  $49/6 \approx 8.17$ . The gap rule ( $c = 2/3$ ) gives  $g^* = (5/3, 1/3)$  and  $42/5 = 8.4$ .*

## B. LOTTERIES AND EQUAL DIVISION

Fix a round, a budget  $B > 0$ , and a pool  $P \subseteq \{1, \dots, n\}$  of  $m$  researchers. The *equal division* over  $P$  grants each researcher in  $P$  the amount  $B/m$ . A *lottery* over  $P$  with  $k \leq m$  winners selects  $k$  researchers from  $P$  uniformly at random and grants each winner  $B/k$ . Expected output is evaluated as in Appendix A: given an allocation  $g$ , it is  $\sum_i \varphi_i(g_i)$ , and for a random allocation, the expectation is taken over the draw as well.

**Proposition 2** (lotteries are outperformed by equal division). *For every pool  $P$ , budget  $B$ , and winner count  $k \leq m$ , the lottery's expected output does not exceed the equal division's, with equality only if  $k = m$ . In particular, a lottery over the whole population, at any winner count, produces at most uniform funding's expected output.*

*Proof.* Researchers outside  $P$  receive nothing under either scheme. For  $i \in P$ , the lottery's grant to  $i$  is a random variable  $\tilde{g}_i$  equal to  $B/k$  with probability  $k/m$  and 0 otherwise, so  $\mathbb{E}[\tilde{g}_i] = B/m$ . By linearity, the lottery's expected output is  $\sum_{i \notin P} \varphi_i(0) + \sum_{i \in P} \mathbb{E}[\varphi_i(\tilde{g}_i)]$ . Since  $\varphi_i$  is strictly concave (Lemma 1(ii)), Jensen's inequality gives  $\mathbb{E}[\varphi_i(\tilde{g}_i)] \leq \varphi_i(\mathbb{E}[\tilde{g}_i]) = \varphi_i(B/m)$  for every  $i \in P$ , with equality only if  $\tilde{g}_i$  is degenerate, that is,  $k = m$ . Summing over  $i$  yields the claim; taking  $P = \{1, \dots, n\}$  yields the comparison with uniform funding.  $\square$

**Remark 1** (what the proposition does and does not cover). *The proposition compares expected output within a fixed pool. It measures the effect of the randomization alone: whatever pool review selects, replacing the lottery within that pool by the equal division of the same money weakly increases expected output. The proposition says nothing about the choice of pool, about the review ranking within the pool that the equal division also ignores, or about anything a lottery saves outside the model (review costs, proposal effort, bias). § 7 measures the first two and cites the literature on the third.*

## C. ROBUSTNESS TO THE PRODUCTION TECHNOLOGY

This appendix asks which of our results depend on the harmonic-mean production function of §3. We embed that function in a family indexed by an exponent  $\gamma$ : expected output is  $A M_\gamma(K, R)$ , where  $M_\gamma(K, R) = ((K^\gamma + R^\gamma)/2)^{1/\gamma}$  is the power mean of capability and resources. At  $\gamma = -1$  the power mean is the harmonic mean  $2KR/(K + R)$ , our form. In the limit  $\gamma \rightarrow 0$  (Cobb-Douglas) it is the geometric mean  $\sqrt{KR}$ , and a shortfall in either input can be offset by more of the other; as  $\gamma$  decreases the inputs grow harder to substitute; in the limit  $\gamma \rightarrow -\infty$  (Leontief)

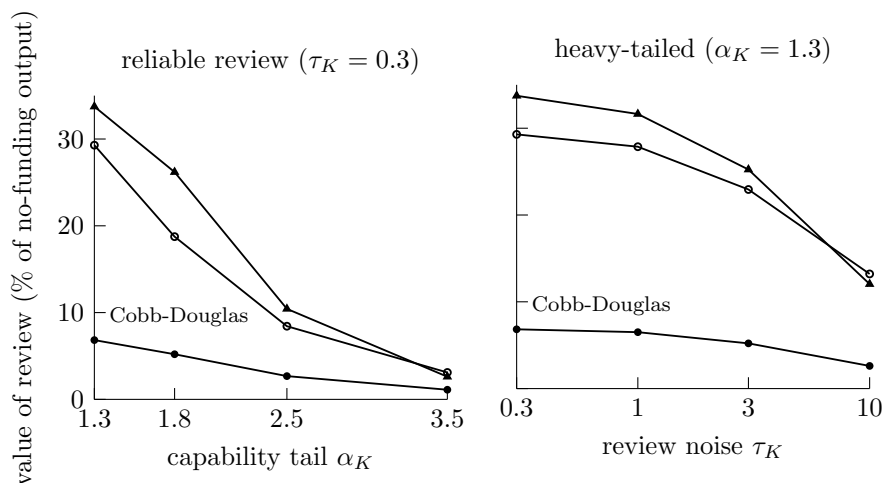


FIGURE 13. The structure of review’s value survives the production function. The value of the review signal as a percent of no-funding output across the family of production functions: at every technology tested, the value increases as capability grows more heavy-tailed (left) and as review grows more informative (right), and its scale increases with complementarity. Filled circles: Cobb-Douglas; open circles:  $\gamma = -3$ ; triangles: Leontief. Settings: Appendix D.

it is  $\min(K, R)$ , and output depends only on the scarcer input. We re-run the model’s central comparisons across this family.<sup>22</sup>

The value of the review signal survives the whole family (Figure 13). At every technology tested, from Cobb-Douglas to Leontief, the signal’s value increases as capability grows more heavy-tailed and as review grows more informative. The ordering of review’s value by field (§5) is therefore not an artifact of our choice of production function. What changes with the technology is the scale. The more complementary the inputs, the more review is worth: under heavy-tailed capability with reliable review, the signal’s value increases from about seven percent of no-funding output at Cobb-Douglas to about thirty percent at  $\gamma = -3$  and a third at Leontief.<sup>23</sup> Complementarity raises what targeting is worth, and with it the value of the information that targeting requires.

The timing results of §6 do not survive the substitutable end of the family (Table 2). Under Cobb-Douglas production the value of scheduling is indistinguishable from zero at every compounding rate, and the optimal schedule stays within about a hundredth of even. At  $\gamma = -3$ , a technology more complementary than

<sup>22</sup> Settings: Appendix D.

<sup>23</sup> 6.8, 29.3, and 33.7 percent of no-funding output; the same ordering holds at every tested tail and noise level at which the value is distinguishable from zero.

TABLE 2. The value of scheduling and the optimal schedule by production technology. Five rounds. The value of scheduling is the forward-looking funder’s gain over the myopic funder, both with review, as a percent of no-funding output; the schedule’s center of mass is 0.5 when spending is even and larger when spending shifts toward later rounds. One hundred simulated populations per point; standard errors of the gain are at most 0.002 (Cobb-Douglas) and 0.09 ( $\gamma = -3$ ) percentage points.

| compounding rate $\epsilon$                               | 0.05   | 0.15   | 0.30  | 0.55  | 0.85  |
|-----------------------------------------------------------|--------|--------|-------|-------|-------|
| <i>Value of scheduling (percent of no-funding output)</i> |        |        |       |       |       |
| Cobb-Douglas ( $\gamma = 0$ )                             | -0.001 | -0.002 | 0.001 | 0.001 | 0.004 |
| $\gamma = -3$                                             | 0.01   | 0.13   | 0.45  | 1.07  | 1.54  |
| <i>Center of mass of the optimal schedule</i>             |        |        |       |       |       |
| Cobb-Douglas ( $\gamma = 0$ )                             | 0.499  | 0.500  | 0.501 | 0.505 | 0.512 |
| $\gamma = -3$                                             | 0.511  | 0.540  | 0.577 | 0.623 | 0.660 |

our form, both the value of scheduling and the shift of spending toward later rounds increase with the compounding rate and exceed their Cobb-Douglas values at every rate. The timing results hold where capability and resources are complements and vanish where they are substitutable.

How widely the informed funder spreads its grants across this family is reported in §7 (Figure 12).

#### D. SIMULATION SPECIFICATION

**Budget and productivity.** Over the whole horizon, the funder’s total budget is  $2bn\mathbb{E}[R_0]$ , where  $\mathbb{E}[R_0]$  is the expected baseline resources of one researcher, so at  $b = 0.5$  the budget equals the community’s expected baseline resources for one round. The budget does not depend on the horizon  $T$ . Strategies that do not plan ahead spend  $1/T$  of the budget each round. For the timing runs of §6 (Figures 8 and 7) the budget is scaled against capability instead,  $2bn\mathbb{E}[K_0]$ , so that a community with few resources does not zero the budget. The simulations set the productivity constant so that expected output is  $K_i R_i / (K_i + R_i)$ , that is,  $A = 1/2$  in the notation of §3.

**Default parameters.** Unless a figure’s settings below say otherwise:  $T = 2$  rounds;  $n = 50$  researchers; capability and baseline resources drawn from Pareto distributions with scale 1 and tail parameters  $\alpha_K = \alpha_R = 2$ ; correlation  $\rho = 0$ ; compounding rate  $\epsilon = 0.1$ ; budget scale  $b = 0.5$ ; review noise  $\tau_K = 1$  and resource-signal noise  $\tau_R = 1$ ; seed-grant fraction 0.25 where seed grants are used.

**Scoring and comparison.** Each strategy is scored by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations; this removes simulation noise from comparisons without favoring any strategy. Every configuration is run on independently drawn populations, with the same random draws (populations, signals, and realized output) used for every strategy, so that comparisons between strategies are paired. Results are reported as each strategy’s improvement over no funding, in percent, unless a figure states another normalization. Standard errors, where shown, are across simulated populations.

**Settings by figure.**

**Code.** The model, the sweeps, and the scripts that produce every figure and number in the paper are in the project repository; the archived version will be cited at publication.

| Figure      | Settings that differ from the defaults                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5           | Review noise $\tau_K$ from 0.05 to 20; $\alpha_K \in \{1.3, 2, 3.5\}$ ; 50 populations per point. The records-only shortfall relative to complete information is 24.6, 11.4, and 3.5 percent of no-funding output at the three tail parameters.                                                                                                                                                                          |
| 4           | $T = 20$ ; 50 populations. Records-only is the myopic funder without review; with review is the myopic funder with review. Under heavy-tailed capability ( $\alpha_K = 1.3$ ) the correlations run 0.07 to 0.43 against 0.92 in round one.                                                                                                                                                                               |
| 6           | True noise $\tau_K = 3$ ; believed noise from 0.1 to 10; $\alpha_K \in \{1.3, 2\}$ ; 200 populations. Each negative value is within two standard errors of zero.                                                                                                                                                                                                                                                         |
| 8           | $T = 6$ ; community with few resources (Pareto scale for resources 0.001); $\alpha_K = 1.3$ ; no review signal ( $\tau_K = 100$ ); $\epsilon = 10^{-4}$ ; budget scaled against capability; $b = 0.5$ (50 populations) and $b = 3$ (24 populations, standard error of each share at most 0.006).                                                                                                                         |
| 7           | $T = 5$ ; forward-looking funder with review; budget scaled against capability; resource scales 0.001, 0.3, and 3; $\epsilon$ from $10^{-4}$ to 0.85; 200 populations. Five resource scales were run; the figure shows three, and the other two lie between them.                                                                                                                                                        |
| 9           | $T$ from 2 to 10 (to 5 for $\epsilon \in \{0.05, 0.15, 0.55\}$ ); $\epsilon \in \{0.05, 0.15, 0.3, 0.55, 0.85\}$ ; 64 to 200 populations per point. The gain is the forward-looking funder's output over the myopic funder's, both with review; negative gains, all within noise of zero, are shown as zero.                                                                                                             |
| 10          | Review-informed forward-looking funder; seed-grant fraction from 0 to 0.75; heavy-tailed field ( $\alpha_K = 1.3$ ) with reliable review ( $\tau_K = 0.3$ ) at $b \in \{0.1, 0.5, 1\}$ , and the default field at $b = 0.5$ ; 200 populations.                                                                                                                                                                           |
| 11          | Single round; $n = 50$ ; a fifth of the population funded; review score $K_i$ plus Gaussian noise of standard deviation $\tau$ ; expected output computed exactly given each allocation; 2,000 populations per point (Monte Carlo standard error at most 0.002). Left: $\alpha_K = 1.3$ , $b = 0.1$ . Right: $\alpha_K = 3.5$ , $b = 0.5$ .                                                                              |
| 13, Table 2 | Production functions with $\gamma \in \{0, -3\}$ and Leontief; 100 populations per point (50 for Leontief); $T = 2$ for the signal value and $T = 5$ for the timing table. These runs use an earlier version of the allocation routine, so their magnitudes are compared with one another and not with the body's figures. Leontief allocations use a finer allocation step, with ties at the kink broken by fill order. |
| 12          | Single round; review-informed funder; $\gamma$ from 0 to $-12$ and Leontief; three field-and-budget combinations ( $\alpha_K = 1.3$ , $b = 0.1$ ; $\alpha_K = 2$ , $b = 1$ ; $\alpha_K = 3.5$ , $b = 1$ ); 200 populations per point, 400 at the refinement points of the evenly spread curve.                                                                                                                           |